

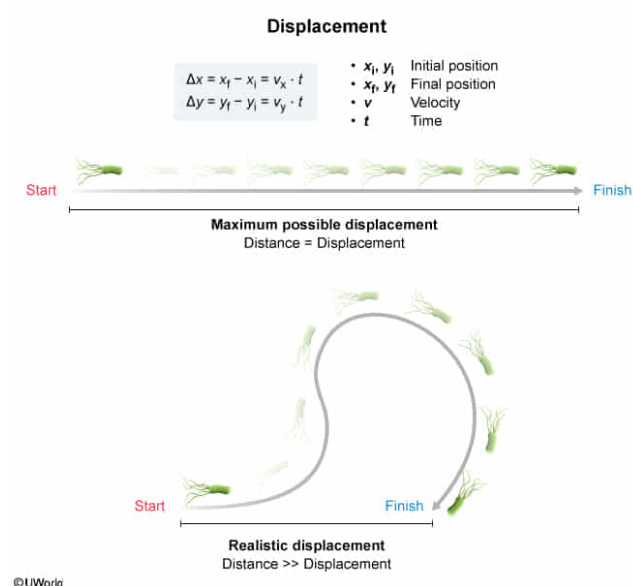
A gram-negative bacterium moves along a flat surface. What is the maximum displacement of the bacterium during a 1-hour period if the bacterium moves at a constant speed of $25 \mu\text{m/s}$?

- ☐ A. $1.5 \times 10^{-3} \text{ m}$ [10%]
- ☐ B. $9.0 \times 10^{-3} \text{ m}$ [19%]
- ☐ C. $1.5 \times 10^{-2} \text{ m}$ [5%]
- ✓ ☒ D. $9.0 \times 10^{-2} \text{ m}$ [64%]

Correct

64% answered correctly

Explanation:



Velocity (v) is a variable of translational motion that expresses the displacement (Δx or $x_f - x_i$) of a moving object per unit **time (t)**. Unlike speed, which is the magnitude of velocity, velocity is a **vector quantity** characterized by both magnitude and direction. Consequently, the speed of an object relates to distance, but the velocity of an object relates to **displacement**.

Due to the relationship between velocity and displacement, the x- and y-components of displacement (Δx and Δy) for an object moving through an x,y-coordinate system can be evaluated by multiplying the x- and y-component velocities (v_x and v_y , respectively) by the interval of time for that motion:

$$\Delta x = v_x \cdot t$$

$$\Delta y = v_y \cdot t$$

In this question, a gram-negative bacterium moves along a flat surface at a constant speed of $25 \mu\text{m/s}$ for 1 hour. Although the bacterium likely moves across the surface along a winding path, the maximum possible displacement from the origin would occur if the movement of the bacteria traces a straight path. In such a case, the constant speed of $25 \mu\text{m/s}$ would translate to a constant velocity of $25 \mu\text{m/s}$ because the bacterium moves in only one direction.

After selecting an x,y-coordinate system oriented such that the motion of the bacterium takes place along only the x-axis (no y-axis displacement, $v_y = 0$), the maximum possible displacement of the bacterium over a span of 1 hour can be determined by substituting velocity and time into the equation for displacement:

$$\Delta x = 25 \frac{\mu\text{m}}{\text{s}}$$

Converting all values to standard units results in:

$$\Delta x = 25 \frac{\mu\text{m}}{\text{s}} \times 3600 \frac{\text{s}}{\text{h}} = 9 \times 10^4 \mu\text{m}$$

Therefore, the maximum hourly displacement of a bacterium moving along a flat surface at a constant speed of $25 \mu\text{m/s}$ is $9 \times 10^4 \mu\text{m}$ (**Choice D**).

Educational objective:

The displacement of an object is calculated as the average velocity of the object multiplied by the interval of time during which the object moves. The speed of an object is equal to the magnitude of the velocity of the object.

Time spent: 0 sec

Subject:

Physics

QID: 402432

System:

Mechanics and Energy